

Solution: $f: \mathbb{R} \rightarrow \mathbb{R}^+ \cup \{0\}$.

$$-2, 2 \in \mathbb{R}. f(-2) = 1 - 2 = -1 \quad f(2) = |2| = 2$$

But $2 \neq -2$. By Defn. 5, f is not one-to-one. By Defn. 8, f is not a bijection.

By Defn. 9, f has no inverse.

$$f: \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}^+ \cup \{0\}.$$

Let, $x, y \in \mathbb{R}^+ \cup \{0\}$ and $f(x) = f(y) \therefore |x| = |y| \Rightarrow x = y$ (By defn. of $|\cdot|$)

By Defn. 5, f is one-to-one.

Let, $f(x) = z$, where $z \in \mathbb{R}^+ \cup \{0\}$. $\therefore |x| = z$. If $x \geq 0$, then $x = z$

If $x < 0$, then $-x = z \Rightarrow x = -z$. Here $x \notin \mathbb{R}^+ \cup \{0\}$.

$\therefore x = z \in \mathbb{R}^+ \cup \{0\}$. By Defn. 7, f is onto.

By Defn. 8, f has a bijection. By Defn. 9, f has an inverse. $\square$

33) Suppose that g is a function from A to B and f is a function from B to C .

a) Show that if both f and g are one-to-one functions, then $f \circ g$ is also one-to-one.

Solution: By Defn. 5, if $x, y \in A$ and $f(x) = f(y)$, then $x = y$.

By Defn. 5, if $a, b \in A$ and $g(a) = g(b)$, then $a = b$.

$$\text{Let, } \exists x, y \in A \text{ s.t. } f(g(x)) = f(g(y))$$

By Defn. 5, $g(x) = g(y)$. By Defn. 5, $x = y$.

$\therefore$ From Defn. 5, $f \circ g$ is also one-to-one. $\square$

b) Show that if both f and g are onto functions, then $f \circ g$ is also onto.

Solution: By Defn. 7, $\forall z \in C, \exists y \in B$ s.t. $f(y) = z$.

By Defn. 7, $\forall x \in B, \exists y \in A$ s.t. $g(y) = x$.

Let, $f(g(x)) = y$ where $y \in C$. By Defn. 7, $g(x) \in B$. By Defn. 7, $x \in A$.

$\therefore$ From Defn. 7, $\forall y \in C, \exists x \in A$ s.t. $f(g(x)) = y$, $f \circ g$ is also onto. $\square$